Your multisig safe with MetaMask

For board members who approve with MetaMask on a computer

mvxsafe.io works with MetaMask through the **MultiversX Snap**, an add-on that MultiversX publishes for MetaMask. The safe and its rules stay the same: nothing leaves it without the signatures it needs. This guide shows what you see at each step, in mvxsafe and in MetaMask. The video that comes with it shows the same steps.

Before you start

- **A computer.** MetaMask in Chrome, the browser this guide was made with. On a phone mvxsafe offers no MetaMask.
- **Your MultiversX address.** Through the Snap, MetaMask gives you a MultiversX address of its own (`erd1…`), not your Ethereum one. Once you are connected, mvxsafe shows it at the top right. The board uses that address for you. It also pays the network fees.
- **A little EGLD on it.** About 0.1 EGLD if you create a safe (that costs about 0.072). A signature costs about 0.0001.
- **Your Secret Recovery Phrase.** That MultiversX address comes from MetaMask's phrase. Restored in another MetaMask, the phrase gives the same address and the same seat on the board. Keep it as you keep any key to company funds. If it is lost, the other board members can vote a new address in.

PART 1

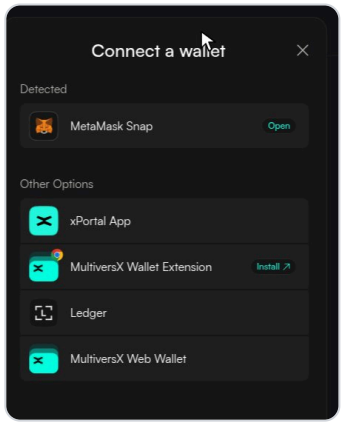
Connect, the first time

MetaMask adds the MultiversX Snap once. Later visits ask only for the last screen.

1 Choose MetaMask Snap

Go to **mvxsafe.io** and press **Connect**. With MetaMask installed, it is listed under **Detected**. Press **MetaMask Snap**.

mvxsafe then shows "Requesting Connection" while MetaMask asks you, in its side panel.

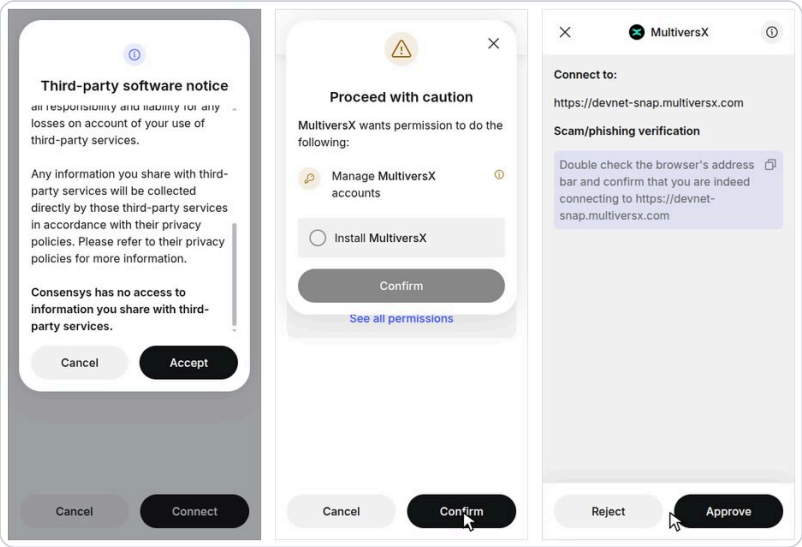


2 Let MetaMask add the Snap

- **Third-party software notice:** read it to the end, then **Accept**.
- **Connection request** ("wants to use @multiversx/metamask-snap"): **Connect**.
- **Add to MetaMask**, with the permissions it asks for: **Confirm**.
- **Proceed with caution:** tick **Install MultiversX**, then press **Confirm** in that same window. The Confirm behind it only closes the window and starts over.
- **Installation complete: OK**.

3 Approve the connection

MetaMask shows "Connect to: https://snap.multiversx.com". That is the Snap's own page, which mvxsafe opens for MetaMask, so this is expected. Your address bar must still read mvxsafe.io. Press **Approve**. You are signed in for 24 hours.



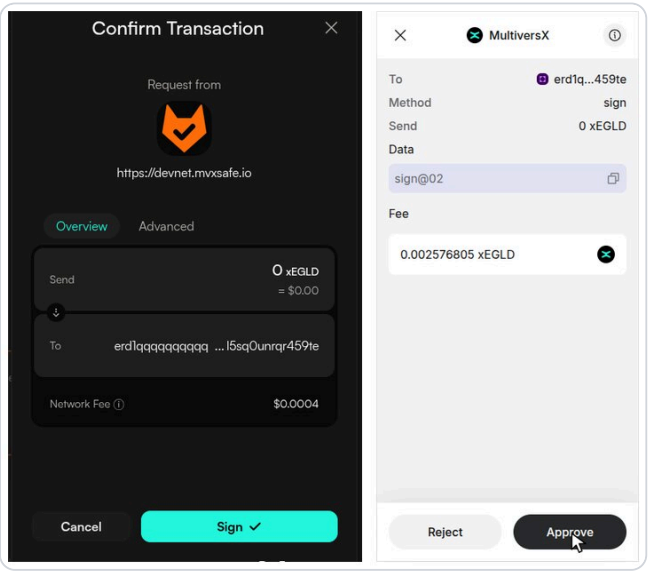
PART 2

Every action: Sign, then Approve

Two presses each time: one in mvxsafe, one in MetaMask.

4 Sign in mvxsafe, then Approve in MetaMask

- mvxsafe opens its **Confirm Transaction** panel: who asks (mvxsafe.io), what your wallet sends (0 EGLD: the safe pays out, never your wallet) and the fee. Press **Sign**.
- MetaMask then shows the call itself: **To** (the safe, its address cut short), **Method**, **Send**, **Data** and **Fee**. **Send must read 0.** Press **Approve**. If anything is not what you expect, press **Reject**: nothing is sent.
- MetaMask's fee is an upper limit, a few thousandths of an EGLD for a signature. In our test the network charged 14 to 44 times less, about 0.0001 EGLD. A fee far above a few thousandths is wrong: press **Reject**.



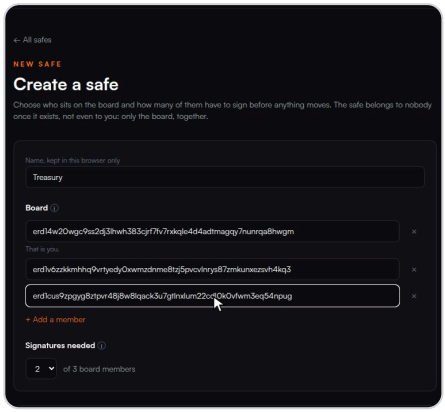
PART 3

Create a safe

Only the person who creates it does this. Everyone else starts at part 4.

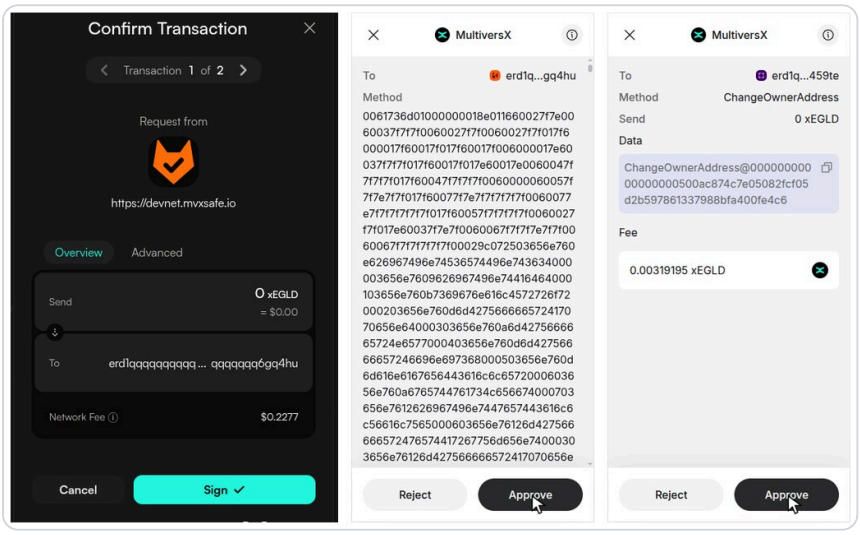
5 Fill in the create page

Press **Create a safe**. Give it a name, kept in your browser only. Your address is the first member. Add each board member's MultiversX address and choose **Signatures needed**, for example 2 of 3. Press **Create a 2 of 3 safe**.



6 Two transactions, each Sign then Approve

- **1 of 2 creates the safe.** MetaMask shows a very long text in hex under **Method**, sent to `erd1qqq…6gq4hu`. That is the safe's program and the network's address for new contracts: expected. The fee is about 0.071 EGLD.
- **2 of 2 hands the safe to itself:** `ChangeOwnerAddress`. From then on even its code needs the board's signatures.



PART 4

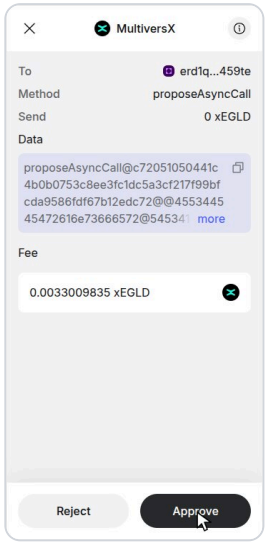
Propose, sign, carry out

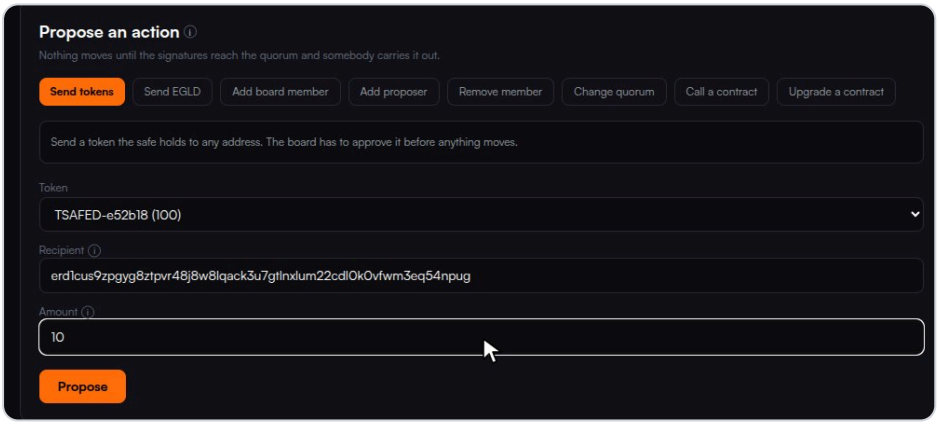
The same as with any wallet, with one thing to check in MetaMask.

7 Propose a payment

On the safe's page, under **Propose an action**, choose **Send tokens**, the token, the recipient and the amount, then **Propose**, Sign and Approve.

MetaMask shows `proposeAsyncCall`, with the recipient and the amount only in hex. Check them on the form before you press Sign. A proposal by a board member counts as their signature.





8 Sign what a colleague proposed

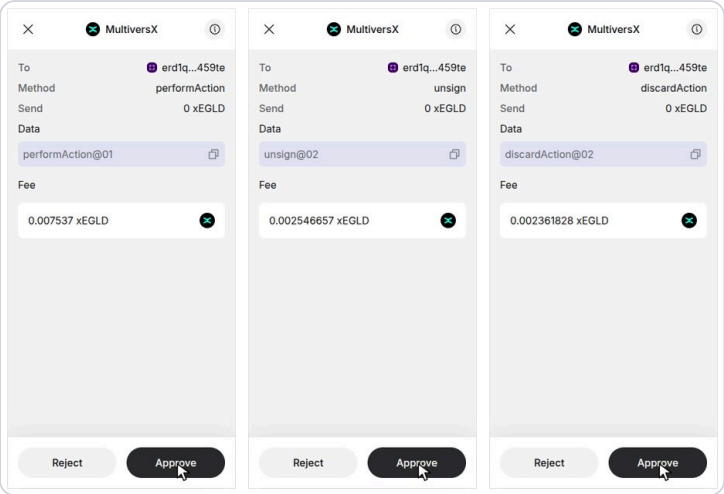
Read the card first: MetaMask does not say what the action does. Then press **Sign**, Sign in the panel and check MetaMask before you Approve:

- **The number.** The card shows `#2 @02`. MetaMask shows `sign@02`: the number in hex. From #10 on they differ: #10 is `@0a`, #16 is `@10`. Match the `@` part.
- **The safe.** Numbers start again in every safe: check that To ends with the same five characters as the safe address at the top of its page.



9 Carry it out, take a signature back or discard

- Once enough have signed, any board member can press **Carry it out**: `performAction@01`.
- Until then **Remove my signature** takes yours back: `unsign@02`.
- When an action has no signatures left, any board member can press **Discard**: `discardAction@02`.



Worth knowing

- Your keys never reach mvxsafe. Every transaction is one you approved in MetaMask.
- If MetaMask shows something you did not ask for, press **Reject**. Nothing is sent.
- The sign-in lasts 24 hours. Then you connect again, with only the last screen of part 1.
- mvxsafe checks every sign-in signature. If it ever says **Sign-in refused**, your wallet did not prove its address: stop and tell the other board members.

The pictures and the video come from the MultiversX test network with test wallets and a test token, so they show a DEVNET badge, xEGLD and the token TSAFED. On mvxsafe.io your safe shows EGLD and FOXSY-5d5f3e. MetaMask then asks to connect to snap.multiversx.com instead of devnet-snap.multiversx.com.